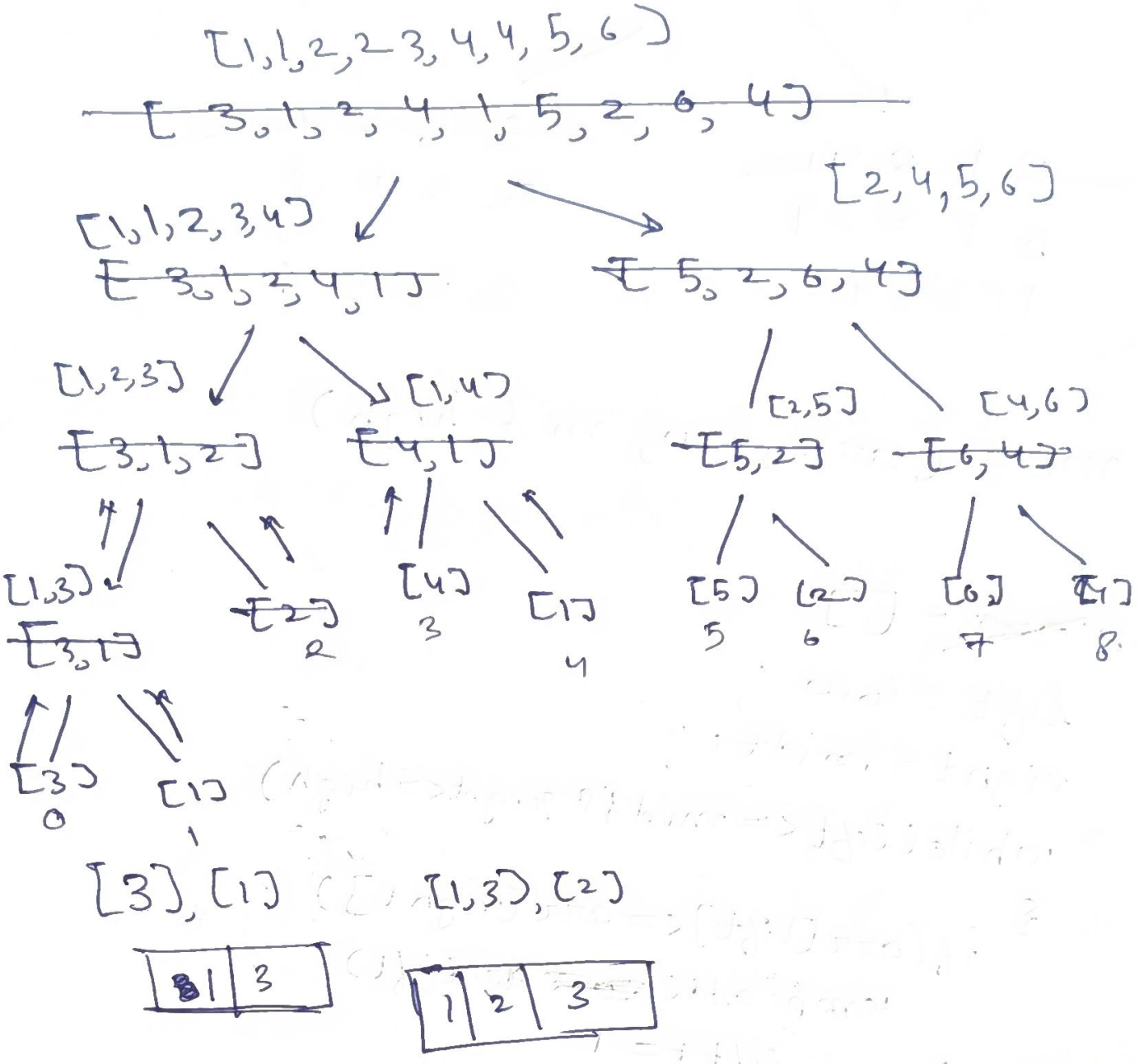
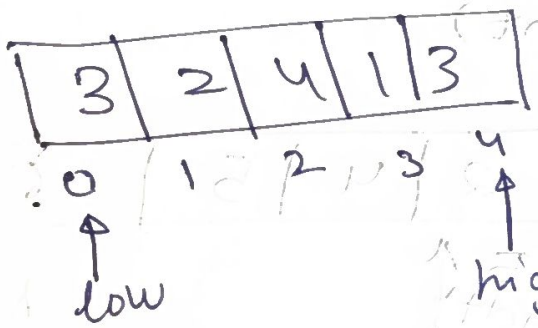
Merge Sort (Recursion)

* Divide and merge



Ex




```
mergeSort(arr, low, high)
{
    mid = (low+high)//2
    mergeSort(arr, low, mid)
    mergeSort(arr, mid, high)
    merge(arr, low, mid, high)
}

if (low >= high)
    return;
```

3, 1, 2, 4, 1, 5, 6, 2, 4
0 1 2 3 4 5 6 7 8

3 1 2 4 1

0 1 2 3 4

1 1 2 3 4

5 6 2 4

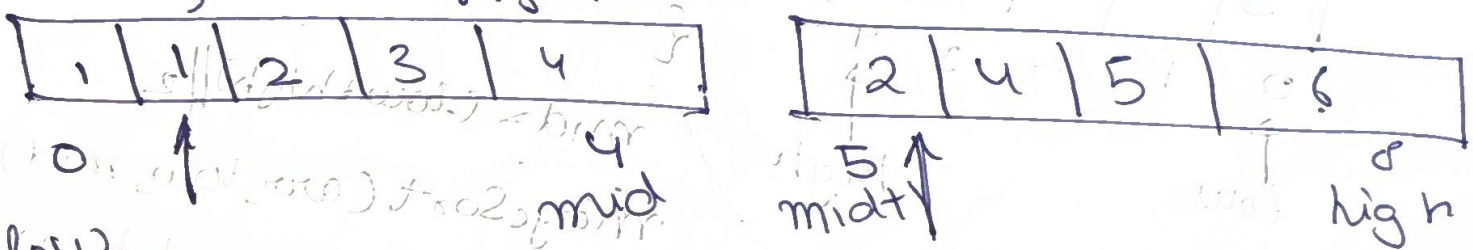
5 6 7 8

2 4 5 6

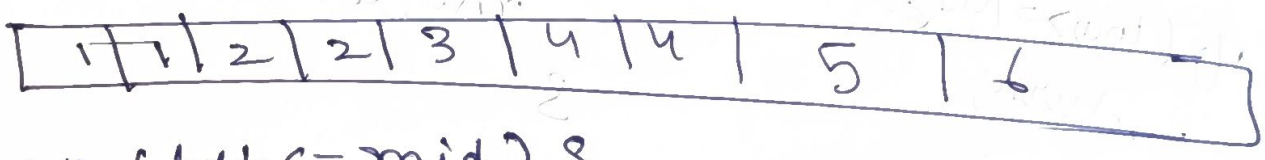
merge(arr, low, mid, high)

temp = []

left = low;
right = mid+1;
while(left <= mid && right <= high)
{
if(arr[left] <= arr[right])
temp.add(arr[left]);
left++;
else
temp.add(arr[right])
right++;



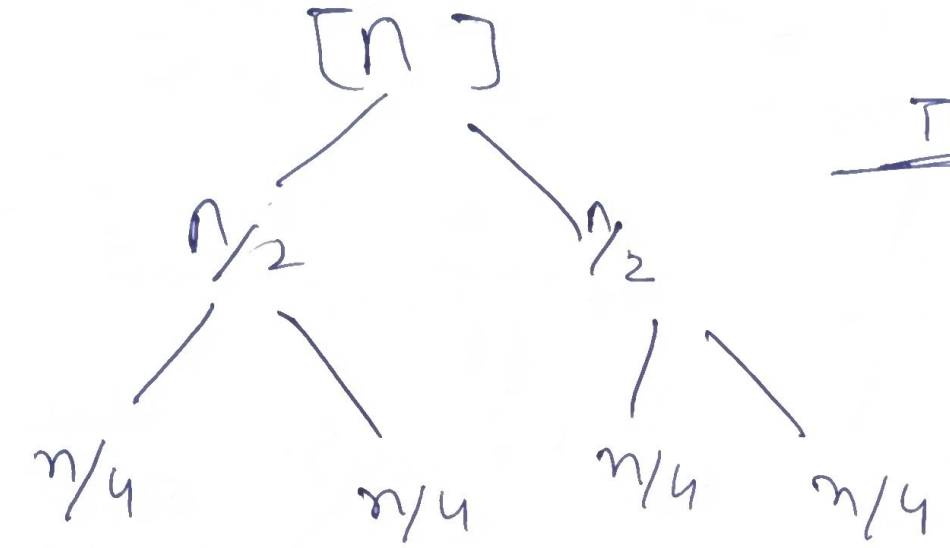
arr[] =



while(left <= mid) {
temp.add(arr[left])
left++;
}
while(right <= high) {
temp.add(arr[right])
right++;
}

```
for (i=low ; i<=high ; i++) {
    arr[i] = temp[i-low];
}
```

add to array elements.



$TC = \log_2 n$